Prateek Mehra

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Summary

Applied Machine Learning Engineer with 4 years of experience taking ML systems from data preparation and model development through cloud/edge deployment, monitoring, and iteration. Experienced in production computer vision, MLOps, data pipelines, model serving, Kubernetes, serverless infrastructure, observability, and retrieval-augmented generation systems.

Experience

- **Eatclub Brands (formerly Box8)** — Bengaluru, India
  Applied Machine Learning Engineer — *June 2022 - Present*
  - **Production ML Systems**: Delivered production ML systems spanning an attendance platform with 1,100 cameras across 550+ outlets, camera-inspection systems covering 7,000+ cameras, and photo-audit workflows across 10,000 Android devices.
  - **Vision Pipelines**: Developed, deployed, and monitored eight computer-vision pipelines processing 100k+ images per day across batch and serverless deployments; reduced manual audit effort by 75% and achieved up to 0.947 mAP@0.5.
  - **Attendance Platform**: Built a low-cost attendance-verification platform that reduced proxy attendance from 22% to 2% and saved USD 20k in operational costs.
  - **Speech Recognition**: Built a Whisper ASR pipeline on AWS GPU nodes to transcribe customer-care calls and extract order IDs; ran 10–13× faster than CPU, with 98% order-ID extraction accuracy on a 147-call test set.
  - **Real-Time Serving**: Built a serverless inference platform using Cloud Functions, Cloud Tasks and Cloud Run (FastAPI/Uvicorn), with scheduler-driven autoscaling, a kill switch and cost controls.
  - **Foundation Model Evaluation**: Benchmarked 15 open-source and commercial models on customer feedback; achieved 100% precision and 80% recall, improving recall 30% over baseline at 810 ms average latency and $0.44 per 10K texts.
  - **MLOps CI/CD**: Standardized 15 model repositories on a shared GitLab CI pipeline covering linting, tests and security checks; reduced combined image size by 86% and reached 100% unit and integration test coverage.
  - **Leadership**: Led project delivery across the vision platform: planned the team's quarters, interviewed SDE-1 and annotator candidates, and mentored five junior ML engineers.

Projects

ECPedia
Cross Repository Q&A Assistant

- **RAG/Hybrid Retrieval**: Built a Slack code Q&A assistant indexing 123k+ code chunks across 16 repositories, combining Jina code embeddings, pgvector and BM25; reached 0.90 Hit@10 and 0.78 MRR on a 130-question evaluation set.
- **Production LLM Service**: Served answers to 20 engineers across 26 Slack channels, with 2.0-second median latency and approximately $0.001 cost per question; added secret redaction and prompt-injection guards.

AI Attendance Platform
Distributed Edge Computer Vision Attendance System

- **Edge ML System**: Designed an offline-first ML architecture with SQLite-backed configuration, RTSP frame buffering, asynchronous processing queues, and ELK observability. Decoupled RTSP frame capture from inference with a FIFO queue, reducing broken-stream rate from 1% to 0.1%.
- **Production Scale**: Scaled the platform to 550+ outlets across 8 Indian cities, processing 11,200 check-ins per day; 97.6% of check-ins were completed through computer vision, with QR as fallback.

Education

- **Motilal Nehru National Institute of Technology (MNNIT Allahabad)** — Prayagraj, India
  Bachelor of Technology; CPI: 8.07 — *July 2018 - June 2022*

Skills

- Frameworks: PyTorch, TensorFlow, Scikit-Learn, OpenCV, Hugging Face Transformers, FastAPI, Streamlit
- ML Systems: Model Training, Model Serving, ML Pipelines, MLOps, RAG, LLM Orchestration
- Languages: Python, C++, Go, SQL (Postgres, SQLite)
- Data/Retrieval: NumPy, Pandas, Airflow, pgvector, ANN Search
- Inference: ONNX, OpenVINO, Quantization, RTSP, Edge AI
- DevOps: Kubernetes, Docker, GitLab CI, Linux, Conda
- Monitoring/Infra: Grafana, Kibana, Elasticsearch, Cloud Logging
- Cloud: GCP (GKE, Cloud Run, Cloud Functions, Compute Engine, Cloud Storage), AWS (EC2, EKS, ECR, S3)